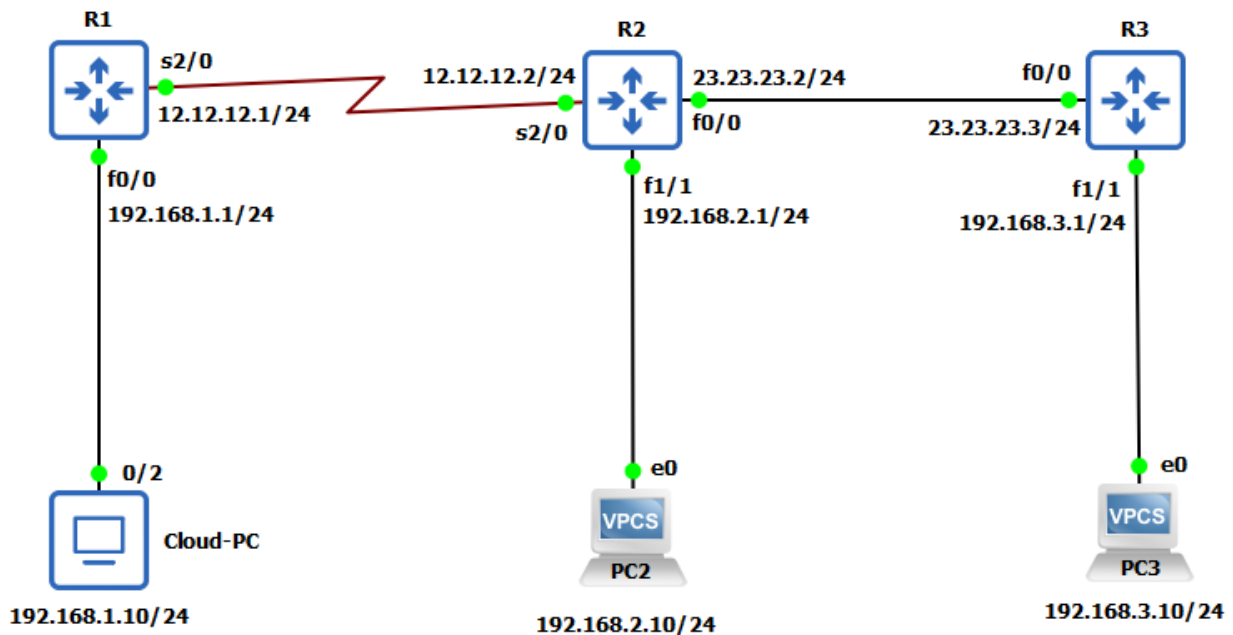


3 Routers Initial Configuration – IPv4



Task

1. Configure routers with IP address as shown in topology
2. Configure enable password as **ccna** and device access password as **cisco**
3. Configure 16 remote access sessions for routers.
4. Remote session should log off after 1 min of idle timeout.
5. Configure PC with IP address and default gateway.

Note: Cloud-PC is a cloud which bridge the R1 router with loopback adapter on real PC.

R1 Router Configuration:

```

R1#enable
R1#config t
R1(config)#hostname R1
R1(config)#no ip domain-lookup
R1(config)#enable password ccna
R1(config)#int fa0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#description Link to local LAN
R1(config-if)#no shut
R1(config-if)#exit
R1(config)#int s2/0
R1(config-if)#ip address 12.12.12.1 255.255.255.0
R1(config-if)#description Link to R2 on s2/0
R1(config-if)#encapsulation hdlc
R1(config-if)#clockrate 64000
R1(config-if)#no shut
  
```

```

R1(config-if)#exit
R1(config)#line vty 0 15
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 1 0
R1(config-line)#exit
R1(config)#line console 0
R1(config-line)#password cisco1
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 0 0
R1(config-line)#exit
R1(config)#exit
R1#copy run start
Destination filename [startup-config]?
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R1#

```

R2 Router Configuration:

```

R2#enable
R2#config t
R2(config)#hostname R2
R2(config)#no ip domain-lookup
R2(config)#enable password ccna
R2(config)#int fa1/1
R2(config-if)#ip address 192.168.2.1 255.255.255.0
R2(config-if)#description Link to local LAN
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int s2/0
R2(config-if)#ip address 12.12.12.2 255.255.255.0
R2(config-if)#description Link to R1 on s2/0
R2(config-if)#encapsulation hdlc
R2(config-if)#clockrate 64000
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#int fa0/0
R2(config-if)#ip address 23.23.23.2 255.255.255.0
R2(config-if)#description Link to R3 on fa0/0
R2(config-if)#no shut
R2(config-if)#exit
R2(config)#line vty 0 15
R2(config-line)#password cisco
R2(config-line)#login
R2(config-line)#logging synchronous

```

```
R2(config-line)#exec-timeout 1 0
R2(config-line)#exit
R2(config)#line console 0
R2(config-line)#password cisco1
R2(config-line)#login
R2(config-line)#logging synchronous
R2(config-line)#exec-timeout 0 0
R2(config-line)#exit
R2(config)#exit
R2#copy run start
Destination filename [startup-config]?
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R2#
```

R3 Router Configuration:

```
R3#enable
R3#config t
R3(config)#hostname R3
R3(config)#no ip domain-lookup
R3(config)#enable password ccna
R3(config)#int fa1/1
R3(config-if)#ip address 192.168.3.1 255.255.255.0
R3(config-if)#description Link to local LAN
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#int fa0/0
R3(config-if)#ip address 23.23.23.3 255.255.255.0
R3(config-if)#description Link to R2 on fa0/0
R3(config-if)#no shut
R3(config-if)#exit
R3(config)#line vty 0 15
R3(config-line)#password cisco
R3(config-line)#login
R3(config-line)#logging synchronous
R3(config-line)#exec-timeout 1 0
R3(config-line)#exit
R3(config)#line console 0
R3(config-line)#password cisco1
R3(config-line)#login
R3(config-line)#logging synchronous
R3(config-line)#exec-timeout 0 0
R3(config-line)#exit
R3(config)#exit
R3#copy run start
Destination filename [startup-config]?
Warning: Attempting to overwrite an NVRAM configuration previously written
```

by a different version of the system image.
 Overwrite the previous NVRAM configuration?[confirm]
 Building configuration...
 [OK]
 R3#

PC2 Configuration:

PC2> ip 192.168.2.10/24 192.168.2.1
 Checking for duplicate address...
 PC1 : 192.168.2.10 255.255.255.0 gateway 192.168.2.1

PC2> **save**
 Saving startup configuration to startup.vpc

PC3 Configuration:

PC3> ip 192.168.3.10/24 192.168.3.1
 Checking for duplicate address...
 PC1 : 192.168.3.10 255.255.255.0 gateway 192.168.3.1

PC3> **save**
 Saving startup configuration to startup.vpc

✓ Verification & Testing:

R1#sh ip int brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.1.1	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
FastEthernet1/0	unassigned	YES	unset	administratively down	down
FastEthernet1/1	unassigned	YES	unset	administratively down	down
Serial2/0	12.12.12.1	YES	manual	up	up
Serial2/1	unassigned	YES	unset	administratively down	down
Serial2/2	unassigned	YES	unset	administratively down	down
Serial2/3	unassigned	YES	unset	administratively down	down

R2#sh ip int brief | exclude unassign

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	23.23.23.2	YES	manual	up	up
FastEthernet1/1	192.168.2.1	YES	manual	up	up
Serial2/0	12.12.12.2	YES	manual	up	up

R3#sh ip int brief | exclude unassign

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	23.23.23.3	YES	manual	up	up
FastEthernet1/1	192.168.3.1	YES	manual	up	up

R1#sh int description

Interface	Status	Protocol	Description
Fa0/0	up	up	Link to local LAN
Fa0/1	admin down	down	
Fa1/0	admin down	down	
Fa1/1	admin down	down	
Se2/0	up	up	Link to R2 on s2/0
Se2/1	admin down	down	
Se2/2	admin down	down	
Se2/3	admin down	down	

R2#sh int description | exclude down

Interface	Status	Protocol	Description
Fa0/0	up	up	Link to R3 on fa0/0
Fa1/1	up	up	Link to local LAN
Se2/0	up	up	Link to R1 on s2/0

R3#sh int description | exclude down

Interface	Status	Protocol	Description
Fa0/0	up	up	Link to R2 on fa0/0
Fa1/1	up	up	Link to local LAN

R1#sh run

Building configuration...

Current configuration : 1762 bytes

!

! Last configuration change at 16:27:09 UTC Mon Oct 1 2020

upgrade fpd auto

version 15.2

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

!

hostname R1

!

boot-start-marker

boot-end-marker

!

!

enable password ccna

!

no aaa new-model

no ip icmp rate-limit unreachable

!

!

no ip domain lookup

ip cef

no ipv6 cef

!

multilink bundle-name authenticated

```
!  
redundancy  
!  
ip tcp synwait-time 5  
!  
interface FastEthernet0/0  
description Link to local LAN  
ip address 192.168.1.1 255.255.255.0  
duplex auto  
speed auto  
!  
interface FastEthernet0/1  
no ip address  
shutdown  
duplex auto  
speed auto  
!  
interface FastEthernet1/0  
no ip address  
shutdown  
duplex auto  
speed auto  
!  
interface FastEthernet1/1  
no ip address  
shutdown  
duplex auto  
speed auto  
!  
interface Serial2/0  
description Link to R2 on s2/0  
ip address 12.12.12.1 255.255.255.0  
serial restart-delay 0  
clock rate 64000  
!  
interface Serial2/1  
no ip address  
shutdown  
serial restart-delay 0  
!  
interface Serial2/2  
no ip address  
shutdown  
serial restart-delay 0  
!  
interface Serial2/3  
no ip address  
shutdown  
serial restart-delay 0  
!  
ip forward-protocol nd
```

```
no ip http server
no ip http secure-server
!
no cdp log mismatch duplex
!
control-plane
!
```

```
mgcp profile default
!
gatekeeper
shutdown
!
```

```
line con 0
exec-timeout 0 0
privilege level 15
password cisco1
logging synchronous
login
stopbits 1
```

```
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
stopbits 1
```

```
line vty 0 4
exec-timeout 1 0
password cisco
logging synchronous
login
transport input all
```

```
line vty 5 15
exec-timeout 1 0
password cisco
logging synchronous
login
transport input all
```

```
!
end
```

R1#

R2#sh run | section interface

```
interface FastEthernet0/0
description Link to R3 on fa0/0
ip address 23.23.23.2 255.255.255.0
duplex auto
speed auto
interface FastEthernet0/1
no ip address
```

```

shutdown
duplex auto
speed auto
interface FastEthernet1/0
no ip address
shutdown
duplex auto
speed auto
interface FastEthernet1/1
description Link to local LAN
ip address 192.168.2.1 255.255.255.0
duplex auto
speed auto
interface Serial2/0
description Link to R1 on s2/0
ip address 12.12.12.2 255.255.255.0
serial restart-delay 0
clock rate 64000
interface Serial2/1
no ip address
shutdown
serial restart-delay 0
interface Serial2/2
no ip address
shutdown
serial restart-delay 0
interface Serial2/3
no ip address
shutdown
serial restart-delay 0

```

R2#sh run | section line

```

line con 0
exec-timeout 0 0
privilege level 15
password cisco1
logging synchronous
login
stopbits 1
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
stopbits 1
line vty 0 4
exec-timeout 1 0
password cisco
logging synchronous
login
transport input all
line vty 5 15

```



```
exec-timeout 1 0
```

```
password cisco  
logging synchronous  
login  
transport input all
```

```
R3#sh run | section interface|line
```

```
interface FastEthernet0/0  
description Link to R2 on fa0/0  
ip address 23.23.23.3 255.255.255.0  
duplex auto  
speed auto  
interface FastEthernet0/1  
no ip address  
shutdown  
duplex auto  
speed auto  
interface FastEthernet1/0  
no ip address  
shutdown  
duplex auto  
speed auto  
interface FastEthernet1/1  
description Link to local LAN  
ip address 192.168.3.1 255.255.255.0  
duplex auto  
speed auto  
interface Serial2/0  
no ip address  
shutdown  
serial restart-delay 0  
interface Serial2/1  
no ip address  
shutdown  
serial restart-delay 0  
interface Serial2/2  
no ip address  
shutdown  
serial restart-delay 0  
interface Serial2/3  
no ip address  
shutdown  
serial restart-delay 0  
line con 0  
exec-timeout 0 0  
privilege level 15  
password cisco1  
logging synchronous  
login  
stopbits 1  
line aux 0
```

```

exec-timeout 0 0
privilege level 15
logging synchronous
stopbits 1
line vty 0 4
exec-timeout 1 0
password cisco
logging synchronous
login
transport input all
line vty 5 15
exec-timeout 1 0
password cisco
logging synchronous
login
transport input all
R3#

```

R1#sh int fa0/0

```

FastEthernet0/0 is up, line protocol is up
Hardware is i82543 (Livengood), address is ca01.2088.0008 (bia ca01.2088.0008)
Description: Link to local LAN
Internet address is 192.168.1.1/24
MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Full-duplex, 100Mb/s, 100BaseTX/FX
ARP type: ARPA, ARP Timeout 04:00:00
Last input 00:00:58, output 00:00:07, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  857 packets input, 129571 bytes
    Received 853 broadcasts (0 IP multicasts)
      0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
      0 watchdog
    0 input packets with dribble condition detected
  249 packets output, 26642 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
  251 unknown protocol drops
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out

```

R1#sh int s2/0

Serial2/0 is up, line protocol is up
 Hardware is M4T
 Description: Link to R2 on s2/0
 Internet address is 12.12.12.1/24
 MTU 1500 bytes, BW 1544 Kbit/sec, DLY 20000 usec,
 reliability 255/255, txload 1/255, rxload 1/255
 Encapsulation HDLC, crc 16, loopback not set
 Keepalive set (10 sec)
 Restart-Delay is 0 secs
 Last input 00:00:04, output 00:00:06, output hang never
 Last clearing of "show interface" counters never
 Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
 Queueing strategy: fifo
 Output queue: 0/40 (size/max)
 5 minute input rate 0 bits/sec, 0 packets/sec
 5 minute output rate 0 bits/sec, 0 packets/sec
 234 packets input, 17701 bytes, 0 no buffer
 Received 234 broadcasts (0 IP multicasts)
 0 runs, 0 giants, 0 throttles
 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
 237 packets output, 17278 bytes, 0 underruns
 0 output errors, 0 collisions, 2 interface resets
 0 unknown protocol drops
 0 output buffer failures, 0 output buffers swapped out
 2 carrier transitions DCD=up DSR=up DTR=up RTS=up CTS=up

R1#sh cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
 S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
 D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
R2	Ser 2/0	145	R	7206VXR	Ser 2/0

R2#sh cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
 S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
 D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
R3	Fas 0/0	138	R	7206VXR	Fas 0/0
R1	Ser 2/0	142	R	7206VXR	Ser 2/0

R3#sh cdp neighbors

Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
 S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
 D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID	Local Intrfce	Holdtme	Capability	Platform	Port ID
R2	Fas 0/0	131	R	7206VXR	Fas 0/0

R1#sh version

Cisco IOS Software, 7200 Software (C7200-ADVENTERPRISEK9-M), Version 15.2(4)M5, RELEASE SOFTWARE (fc2)
Technical Support: <http://www.cisco.com/techsupport>
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Fri 13-Sep-13 18:47 by prod_rel_team

ROM: ROMMON Emulation Microcode

BOOTLDR: 7200 Software (C7200-ADVENTERPRISEK9-M), Version 15.2(4)M5, RELEASE SOFTWARE (fc2)

R1 uptime is 38 minutes

System returned to ROM by unknown reload cause - suspect boot_data[BOOT_COUNT] 0x0, BOOT_COUNT 0, BOOTDATA 19
System image file is "tftp://255.255.255.255/unknown"
Last reload reason: unknown reload cause - suspect boot_data[BOOT_COUNT] 0x0, BOOT_COUNT 0, BOOTDATA 19

This product contains cryptographic features and is subject to United States and local country laws governing import, export, transfer and use. Delivery of Cisco cryptographic products does not imply third-party authority to import, export, distribute or use encryption. Importers, exporters, distributors and users are responsible for compliance with U.S. and local country laws. By using this product you agree to comply with applicable laws and regulations. If you are unable to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at: <http://www.cisco.com/wwl/export/crypto/tool/stqrg.html>

If you require further assistance please contact us by sending email to export@cisco.com.

Cisco 7206VXR (NPE400) processor (revision A) with 491520K/32768K bytes of memory.
Processor board ID 4279256517
R7000 CPU at 150MHz, Implementation 39, Rev 2.1, 256KB L2 Cache
6 slot VXR midplane, Version 2.1

Last reset from power-on

PCI bus mb0_mb1 (Slots 0, 1, 3 and 5) has a capacity of 600 bandwidth points.
Current configuration on bus mb0_mb1 has a total of 800 bandwidth points.
The set of PA-2FE, PA-POS-2OC3, and I/O-2FE qualify for "half bandwidth points" consideration, when full bandwidth point counting results in oversubscription, under the condition that only one of the two ports is used. With this adjustment, current configuration on bus mb0_mb1 has a total of 400 bandwidth points.
This configuration is within the PCI bus capacity and is supported under the above condition.

PCI bus mb2 (Slots 2, 4, 6) has a capacity of 600 bandwidth points.

Current configuration on bus mb2 has a total of 0 bandwidth points
This configuration is within the PCI bus capacity and is supported.

Please refer to the following document "Cisco 7200 Series Port Adaptor
Hardware Configuration Guidelines" on Cisco.com <<http://www.cisco.com>>
for c7200 bandwidth points oversubscription and usage guidelines.

4 FastEthernet interfaces
4 Serial interfaces
509K bytes of NVRAM.

8192K bytes of Flash internal SIMM (Sector size 256K).
Configuration register is 0x2102

R1#

Ping default gateway from Cloud-PC

C:\Users\Ratan>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=8ms TTL=255
Reply from 192.168.1.1: bytes=32 time=9ms TTL=255
Reply from 192.168.1.1: bytes=32 time=12ms TTL=255
Reply from 192.168.1.1: bytes=32 time=6ms TTL=255

Ping statistics for 192.168.1.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 6ms, Maximum = 12ms, Average = 8ms

Ping default gateway from PC2

PC2> ping 192.168.2.1

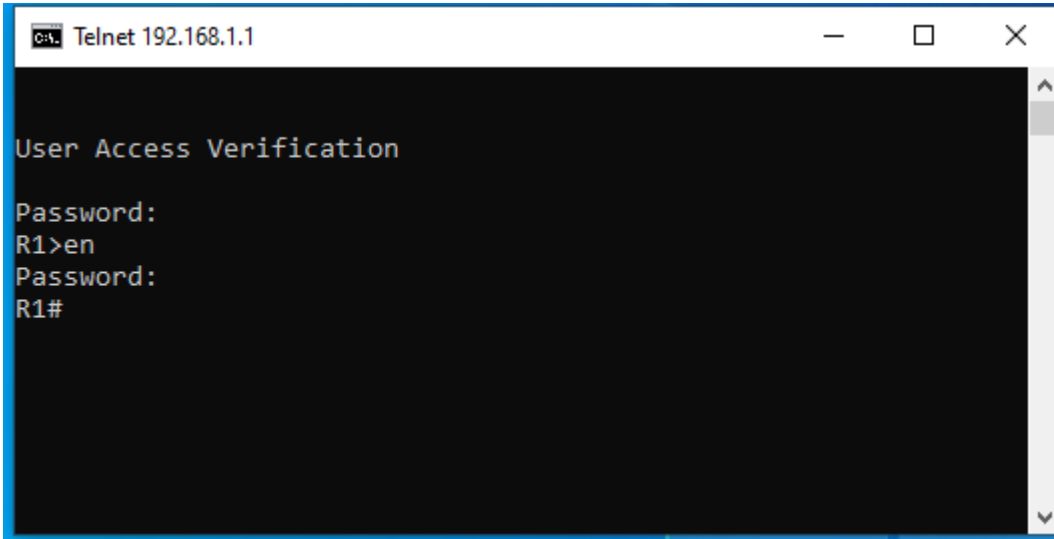
84 bytes from 192.168.2.1 icmp_seq=1 ttl=255 time=31.179 ms
84 bytes from 192.168.2.1 icmp_seq=2 ttl=255 time=7.200 ms
84 bytes from 192.168.2.1 icmp_seq=3 ttl=255 time=7.244 ms
84 bytes from 192.168.2.1 icmp_seq=4 ttl=255 time=3.315 ms
84 bytes from 192.168.2.1 icmp_seq=5 ttl=255 time=5.296 ms

Ping default gateway from PC3

PC3> ping 192.168.3.1

84 bytes from 192.168.3.1 icmp_seq=1 ttl=255 time=21.181 ms
84 bytes from 192.168.3.1 icmp_seq=2 ttl=255 time=11.192 ms
84 bytes from 192.168.3.1 icmp_seq=3 ttl=255 time=9.360 ms
84 bytes from 192.168.3.1 icmp_seq=4 ttl=255 time=9.336 ms
84 bytes from 192.168.3.1 icmp_seq=5 ttl=255 time=9.286 ms

Telnet R1 router from Cloud-PC



```
Telnet 192.168.1.1

User Access Verification

Password:
R1>en
Password:
R1#
```

Important Commands:

```
sh run
sh start
sh version
sh ip int brief
sh ip int brief | exclude unassign
sh int description
sh int description | exclude down
sh run | section interface
sh run | section line
sh run | section interface|line
sh int fa0/0
sh cdp neighbors
copy run start or write
```

Rsatech

Ratan